

1) Project Title

Cash Curve – Design and Development of a Full-Stack Personal Expense Tracker and Financial Insight System

2) Statement of the Problem

Many individuals and small families face difficulty in tracking their daily expenses and understanding where their money is being spent. Most people record expenses manually or do not track them at all, which leads to poor budgeting, unnecessary spending, and lack of financial awareness.

Existing expense tracking applications are either too complex or fail to provide clear insights into spending patterns. There is a need for a simple, user-friendly system that helps users record expenses, analyze spending behavior, and make informed financial decisions.

3) Why Is This Particular Topic Chosen

This topic is chosen because:

- Personal financial management is a **common real-world problem**
- Expense tracking is relevant to students, families, and small businesses
- The project allows strong **Full-Stack implementation**
- It is easy to understand and **practical for BCA level**
- Financial awareness is increasingly important in the digital economy
- The system can be extended with AI-based insights in the future

4) Objective and Scope

Objectives

The objectives of this project are:

- To develop a full-stack web application for tracking expenses
- To categorize and store daily income and expense records
- To analyze spending patterns over time
- To provide visual insights using charts and summaries
- To help users plan budgets and control expenses
- To improve financial discipline and awareness

Scope

The scope of the project includes:

- Daily expense and income entry
- Expense categorization (food, travel, bills, etc.)
- Monthly and yearly financial summaries
- Graphical representation of spending patterns
- Web-based access from any device

Future scope includes AI-based spending prediction and mobile app integration.

5) Methodology

The project follows a structured development approach:

1. **Requirement Analysis**
Understanding user needs and expense tracking workflows.
2. **System Design**
Designing database schema and application architecture.
3. **Frontend Development**
Creating user interfaces for data entry and visualization.
4. **Backend Development**
Handling business logic, data storage, and API integration.
5. **Data Analysis**
Processing expense data to generate summaries and trends.
6. **Testing & Deployment**
Testing functionality and deploying the system.

6) Hardware & Software Requirements

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 8 GB
- Storage: Minimum 256 GB
- Internet connection
- Standard input/output devices

Software Requirements

- Operating System: Windows / Linux
 - Frontend: HTML, CSS, JavaScript, Bootstrap
 - Backend: Python (Django / Flask)
 - Database: MySQL / SQLite
 - Visualization: Chart.js
 - Development Tools: VS Code, Browser
 - Server: Localhost / Cloud Hosting
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7) Contributions of the Project to Society

This project contributes to society by:

- Improving financial awareness among individuals
- Encouraging responsible spending habits
- Helping students and families manage money effectively
- Supporting better budgeting and savings planning
- Promoting digital financial literacy

8) Schedule of the Project

Phase	Activity	Duration
Phase 1	Requirement Analysis	1 Week
Phase 2	System Design	1 Week
Phase 3	Frontend Development	2 Weeks
Phase 4	Backend Development	2 Weeks
Phase 5	Data Analysis & Visualization	2 Weeks
Phase 6	Testing & Debugging	1 Week
Phase 7	Documentation & Submission	1 Week

Total Duration: 10 Weeks

Gantt Chart

[illegible]

9) Limitations of the Project

- Accuracy depends on user-entered data
- Does not integrate with bank APIs in the initial version
- Limited support for multiple users in early implementation
- No real-time financial alerts in the basic version
- Requires regular data entry for effective tracking

10) References

1. Reserve Bank of India – Financial Literacy Resources
2. Django Official Documentation
3. Chart.js Documentation
4. MySQL Official Documentation
5. Research articles on personal finance management
6. Online blogs and journals on budgeting and expense tracking